

## **NOTES – UNIT - 3**

### **Internetworking (Network Layer)**

#### **Introduction**

- The **Network Layer** is part of the communication process in computer networks.
- Its **main job** is to move data packets between **different networks**.
- It ensures packets travel from the **sender to the receiver** through multiple paths.
- The main protocol used here is the **Internet Protocol (IP)**.
- Other network layer protocols handle **routing, testing, and encryption**.

#### **Features of the Network Layer**

1. **Transfers data packets** from source to destination.
2. **Does not modify** the data content.
3. If packets are **too large**, they are **fragmented** (split into smaller packets).
4. Decides the **best route** for data (this is called **routing**).

#### **Services Offered by the Network Layer**

1. **Packetizing**
  - It means **encapsulation and decapsulation** of data.
  - **At sender:** The network layer adds a **header** (with source & destination addresses) to the data and passes it to the data link layer.
  - **At receiver:** The header is removed to get back the original data (payload).
2. **Routing**
  - It is the **process of selecting the best path** to send data from source to destination.
  - Multiple routes may exist — routing decides the **optimal route**.
3. **Forwarding**
  - It is the **actual movement of packets** from one router to another.
  - Each router decides **where to send** the packet next.
  - **Unicast routing:** Packet sent to one destination.
  - **Multicast routing:** Packet sent to multiple destinations.

### **Devices in Internetworking**

**Repeater** – Amplifies signals to extend network range. *(Layer 1)*

**Hub** – Connects devices and sends data to all. *(Layer 1)*

**Bridge** – Connects LAN segments, filters by MAC. *(Layer 2)*

**Switch** – Sends data to specific device using MAC table. *(Layer 2/3)*

**Router** – Connects networks, routes using IP address. *(Layer 3)*

**Gateway** – Connects networks with different protocols. *(All layers)*

**Modem** – Converts digital ↔ analog signals. *(Layer 1)*

**Access Point (AP)** – Provides Wi-Fi connection to wired LAN. *(Layer 2)*

**Firewall** – Controls and secures network traffic. *(Layer 3 and above)*

## Internet Protocol (IP)

### IPv4 – Introduction

- **IPv4** = Internet Protocol version 4. It is a **connectionless protocol** (no dedicated path).
- Uses **32-bit (4-byte)** addressing → gives  **$2^{32}$  unique addresses**.
- IPv4 addresses are written in **dotted-decimal notation** (e.g., 192.168.1.5).
- Divided into **5 classes**: A, B, C, D, and E.

### IPv4 Datagram Header Fields

| Field                         | Description                                                   |
|-------------------------------|---------------------------------------------------------------|
| <b>Version</b>                | Indicates IP version (4 for IPv4).                            |
| <b>HLEN</b>                   | Header length (minimum $5 \times 4$ bytes = 20 bytes).        |
| <b>Type of Service (ToS)</b>  | Defines priority – delay, throughput, reliability.            |
| <b>Total Length</b>           | Size of header + data (max 65,535 bytes).                     |
| <b>Identification</b>         | Unique ID for datagram fragments.                             |
| <b>Flags</b>                  | 3 control bits: Reserved, Don't Fragment, More Fragments.     |
| <b>Fragment Offset</b>        | Position of fragment in original message.                     |
| <b>Time To Live (TTL)</b>     | Limits packet lifetime, prevents looping.                     |
| <b>Protocol</b>               | Identifies next layer protocol (e.g., TCP=6, UDP=17, ICMP=1). |
| <b>Header Checksum</b>        | Detects errors in header.                                     |
| <b>Source IP Address</b>      | Address of sender.                                            |
| <b>Destination IP Address</b> | Address of receiver.                                          |
| <b>Options</b>                | Extra control info like record route or source route.         |

### Types of IPv4 Addresses

#### 1. Public IP Address

- Used on the **internet** (externally visible).
- Assigned by the **Internet Service Provider (ISP)**.
- Two types:
  - **Dynamic IP** – changes each time device connects.
  - **Static IP** – permanent and fixed (used by servers, DNS, etc.).

#### 2. Private IP Address

- Used **inside local networks (LANs)** only.
- **Not accessible** directly from the internet.
- Helps internal communication between devices.
- Defined ranges:
  - **Class A:** 10.0.0.0 – 10.255.255.255
  - **Class B:** 172.16.0.0 – 172.31.255.255
  - **Class C:** 192.168.0.0 – 192.168.255.255

### **Subnet Mask**

- A **32-bit number** that divides IP into **Network** and **Host** parts.
- Identifies which portion of IP is **network** and which is **device**.
- Default subnet masks:
  - Class A → **255.0.0.0**
  - Class B → **255.255.0.0**
  - Class C → **255.255.255.0**

**Dynamic IP** Changes each time

**Static IP** Fixed, permanent

**Subnet Mask** Splits IP into network & host parts

## **VPN and IPsecurity**

### **Definition**

- A **VPN (Virtual Private Network)** creates a **safe and encrypted connection** over the Internet. It allows users to **connect securely** to a private network even from remote locations.
- Uses **tunneling protocols** to keep communication private.

### **Need for VPN**

- Protects **privacy and data security**.
- Allows **secure access** to company networks for remote users.
- **Prevents eavesdropping** and unauthorized access.
- Helps bypass **geo-blocking and censorship**.
- Useful for **safe browsing on public Wi-Fi**.

### **Uses of VPN**

1. **Site-to-Site VPN** – Connects two distant networks (e.g., company branches).
2. **Remote Access VPN** – Allows individual users to connect securely from home or elsewhere.

### **Advantages of VPN**

- Hides **IP address** and encrypts data.
- Provides **secure communication** over public networks.
- Ensures **privacy** and **safe remote work**.

## IP Security (IPSec)

### Definition

- **IPSec** is a **set of security protocols** that protect data transmitted over IP networks.
- Ensures **confidentiality**, **integrity**, and **authenticity** of data.
- Works at the **Network Layer**.

### Main Components

1. **Authentication Header (AH):**
  - Verifies the **source of data**.
  - Provides **authentication** and **integrity**.
2. **Encapsulating Security Payload (ESP):**
  - Provides **encryption**, **authentication**, and **integrity**.
  - Uses algorithms like **AES** (Advanced Encryption Standard) and **3DES**.
3. **Security Association (SA):**
  - Manages **security settings** between devices (negotiates keys, algorithms, etc.).

### How IPSec Works

- Combines **Authentication** + **Encryption** to secure communication.
- Authentication confirms the sender's identity.
- Encryption hides data from unauthorized users.

### Benefits of IPSec

- Protects **all traffic** passing through routers/firewalls.
- Works **below the transport layer (TCP/UDP)** – transparent to applications.
- **No user training** required.
- Supports **secure remote access** for offsite employees.
- Useful for **virtual private networks** and **secure company networks**.

| Concept | Function                                       |
|---------|------------------------------------------------|
| VPN     | Secure private connection over public internet |
| Need    | Privacy, security, remote access               |
| IPSec   | Provides security to data on IP networks       |

| <b>Concept</b> | <b>Function</b>                        |
|----------------|----------------------------------------|
| <b>AH</b>      | Authentication & integrity             |
| <b>ESP</b>     | Encryption, authentication & integrity |
| <b>SA</b>      | Manages encryption setup               |

## Dynamic Host Configuration Protocol (DHCP)

- **DHCP** stands for **Dynamic Host Configuration Protocol**.
- It is used to **automatically assign IP addresses** to devices in a network.
- Works based on the **client-server model**.
- Called a “**plug and play**” or “**zero configuration**” protocol.

### **Need for DHCP**

- Large organizations get address blocks from **ICANN** or **ISP**.
- Instead of assigning IPs **manually**, DHCP automates the process.
- Makes IP management easier, especially in **large or dynamic networks**.

### **Goals of DHCP**

- Allow hosts to **dynamically obtain IP addresses** when joining the network.
- Assigns IP **on a leased basis** (temporary use).
- IP can be **renewed** or **released** when disconnected.
- Enables **reuse of IP addresses**.
- Supports **mobile users** who connect from different locations.

### **DHCP Working (4 Main Steps)**

#### **1. DHCP Discover –**

Host broadcasts a message to find available DHCP servers.

#### **2. DHCP Offer –**

DHCP servers reply with an offer message containing IP details.

#### **3. DHCP Request –**

Host selects one offer and requests that IP address.

#### **4. DHCP Acknowledgment (ACK) –**

Server confirms and assigns the IP address to the host.

*(If not approved, server sends a “DHCP NAK”)*

### **Technical Details**

- DHCP uses **UDP ports 67 (server) and 68 (client)**.
- Operates at the **Application Layer**.
- DHCP messages are **encapsulated in UDP datagrams**.



| Concept       | Description                               |
|---------------|-------------------------------------------|
| Full Form     | Dynamic Host Configuration Protocol       |
| Layer         | Application Layer                         |
| Function      | Automatically assigns IP addresses        |
| Ports Used    | UDP 67 (Server), UDP 68 (Client)          |
| Main Messages | Discover, Offer, Request, ACK             |
| Type          | Plug & Play / Zero Configuration Protocol |
| Lease         | Temporary IP assignment                   |

### **Routing Protocols – RIP (Routing Information Protocol)**

**RIP** stands for **Routing Information Protocol**.

- It is an **Interior Gateway Protocol (IGP)** used **within an autonomous system**.
- RIP is based on a **distance vector routing algorithm**.

#### **Basic Concept**

- Considers the network as a **graph**:
  - **Nodes** → Routers and **Links** → Networks
- Each router maintains a **routing table** to know how to reach every destination.

#### **Metric Used**

- RIP uses **hop count** as its metric.
- **Hop count** = number of routers (or networks) between source and destination.
- **Maximum hops** = **15** (16 = unreachable).
- Hence, RIP is suitable for **small networks**.

#### **Working**

1. Every router **broadcasts its routing table** to its neighbors **every 30 seconds**.
2. Neighbors update their tables and pass it on to others → this is called **convergence**.
3. If a router gets a **shorter route**, it updates its table.

4. If the route is **longer**, it waits through a **hold-down period** before updating.

| Feature       | Description               |
|---------------|---------------------------|
| Protocol Type | Interior Gateway Protocol |
| Algorithm     | Distance Vector           |
| Metric        | Hop Count                 |
| Max Hop       | 15                        |
| Updates       | Every 30 seconds          |
| Suitable For  | Small networks            |
| Example Use   | Small LAN-based networks  |

### **Routing Protocols – OSPF (Open Shortest Path First)**

#### **Definition**

- **OSPF = Open Shortest Path First.**
- It is an **Interior Gateway Protocol (IGP)** used **within an autonomous system**.
- Based on the **Link-State Routing Algorithm**.

#### **Working Principle**

- Each router has full knowledge of the **network topology**.
- Routers exchange information using **Link State Advertisements (LSA)**.
- All LSAs are stored in a **Link State Database (LSDB)**.
- Routers calculate the **shortest path** using Dijkstra's algorithm.

#### **Areas in OSPF**

- OSPF divides the network into **Areas** for easier management.
- Each **Area** contains routers, networks, and hosts.
- **Area Border Routers (ABR):** Routers that connect different areas and summarize information to share between them.

#### **Link State Packet (LSP)**

Contains:

1. ID of the router that created it.

2. Cost of link to each neighbor.
3. Sequence number.
4. Time-to-live (TTL).

Helps in **reliable flooding** of information to all routers.

### Flooding Process

- When a router gets an LSP, it **floods** it to all neighbors except the sender.
- This continues until **all routers** have the same network information.

| Feature       | Description                    |
|---------------|--------------------------------|
| Full Form     | Open Shortest Path First       |
| Protocol Type | Interior Gateway Protocol      |
| Algorithm     | Link State                     |
| Metric        | Cost (based on bandwidth)      |
| Updates       | When changes occur             |
| Works With    | Areas and Routers              |
| Advantage     | Faster, more reliable than RIP |